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Modul 6: Magnetic quantities

Stand: 18. Februar 2026

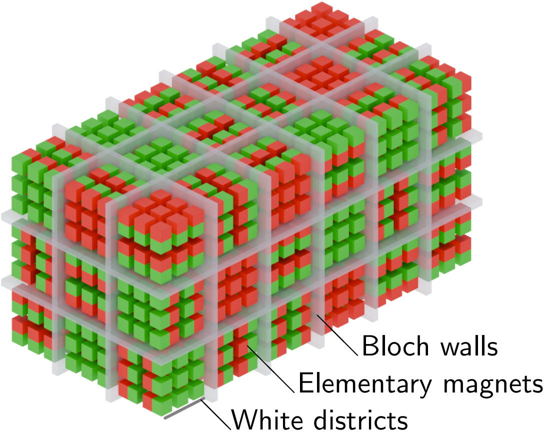
License

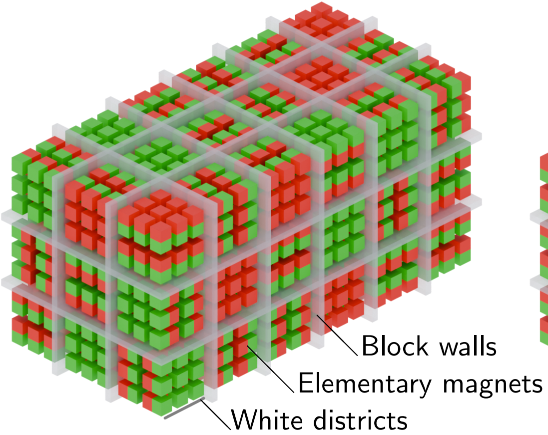
Learning objectives: Magnetic quantities

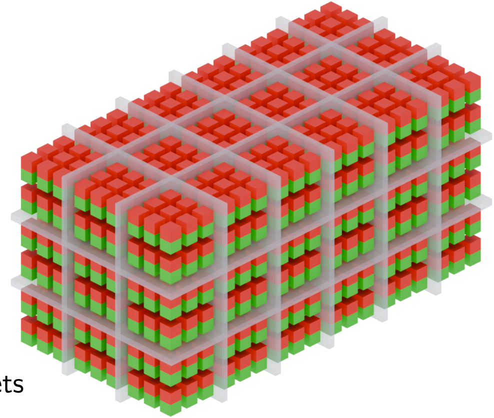
The students

- know the basic magnetic quantities in the magnetic circuit.
- understand the physical principles behind the individual magnetic quantities.
- can describe the interactions between magnetic quantities.
- can calculate the individual quantities in the magnetic circuit.

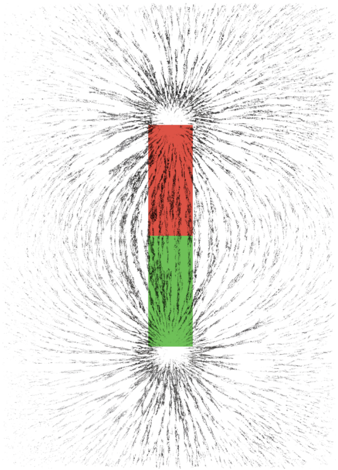
Magnetism




Block walls
Elementary magnets
White districts



Magneitc field



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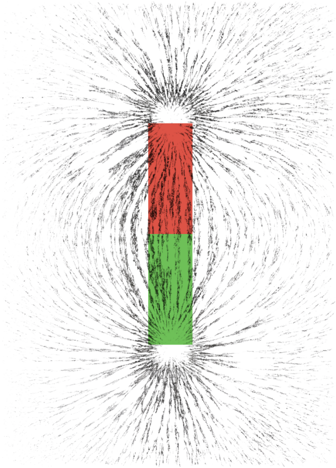

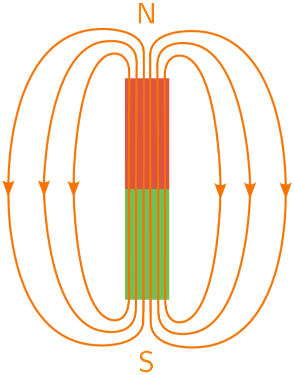

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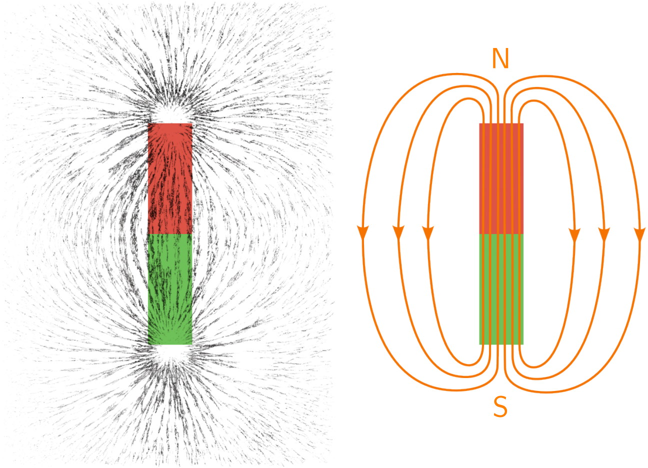
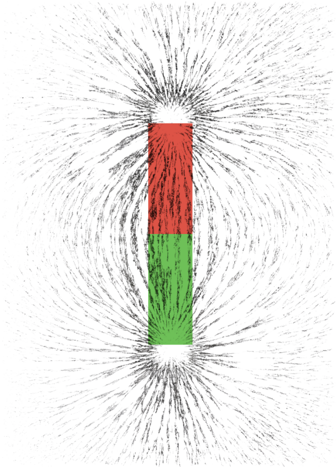
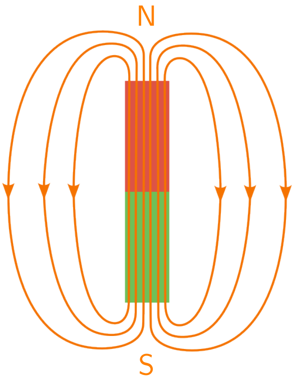


- Magnetic field lines are always closed (source-free)..

Magnetic field





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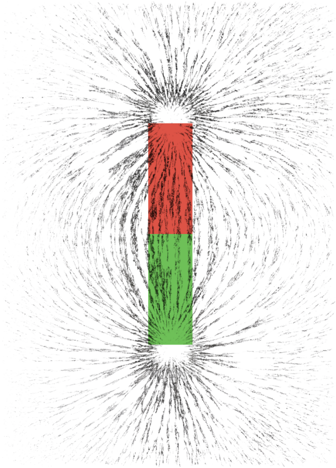
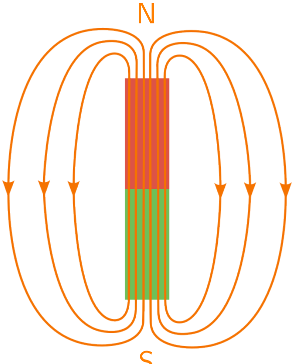
- Outside the magnet, they run from the north pole to the south pole.

Magnetic field





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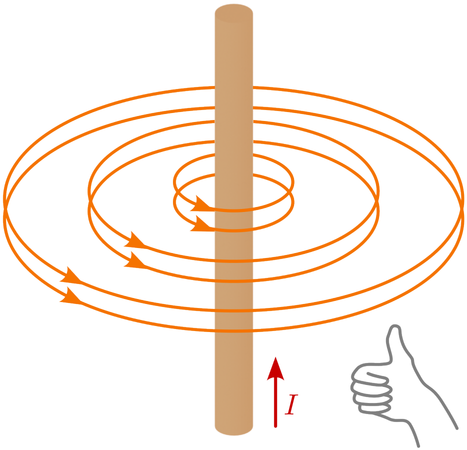
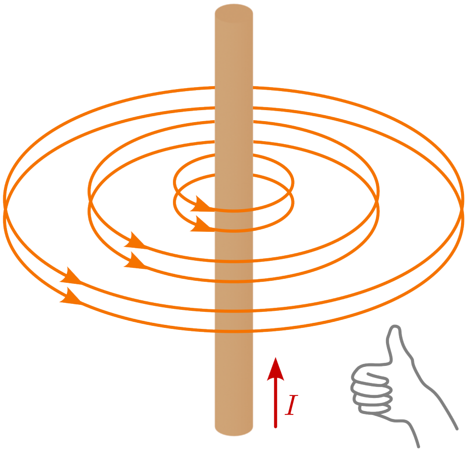
- Outside the magnet, they run from the north pole to the south pole.
- They always exit or enter the magnetic surface perpendicularly.

Electromagnetism

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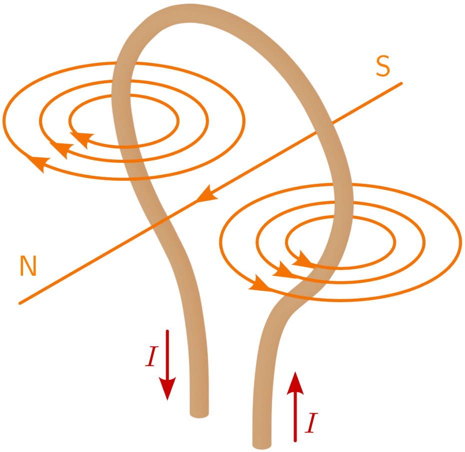
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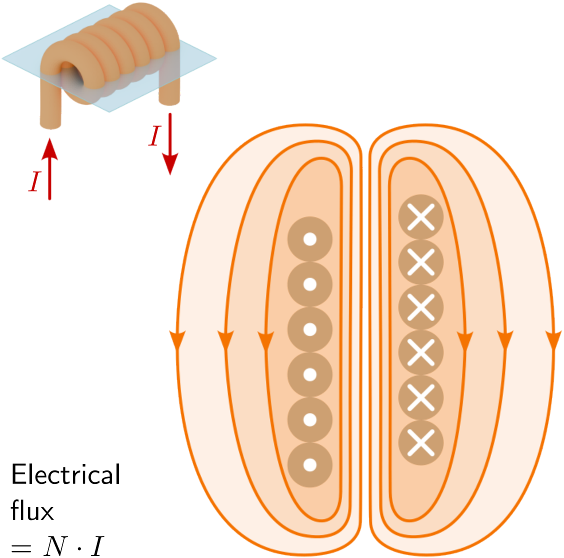
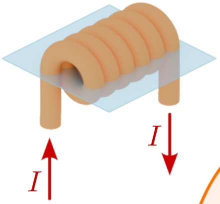


Flux or magnetic tension Θ (Theta) $=$ Total current of a fluxed area

Electrical flux
$= N \cdot I$



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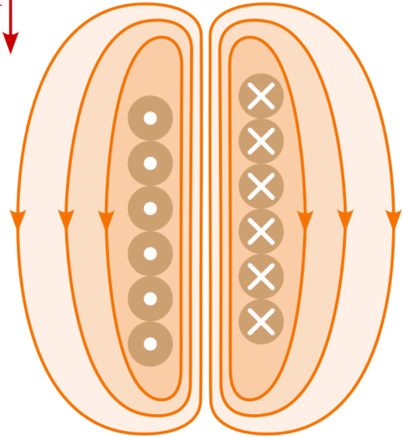

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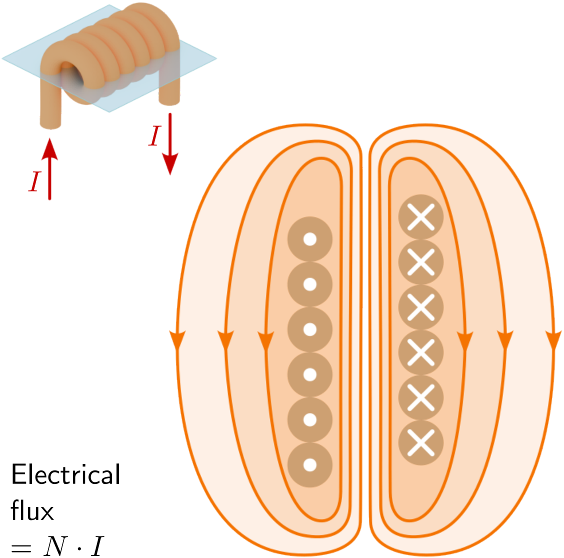

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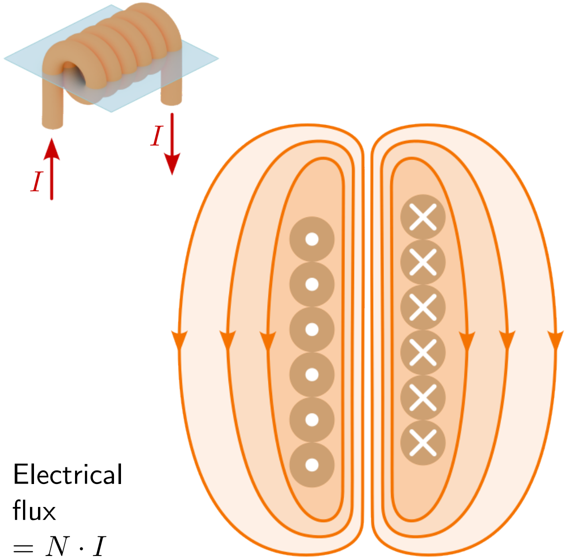

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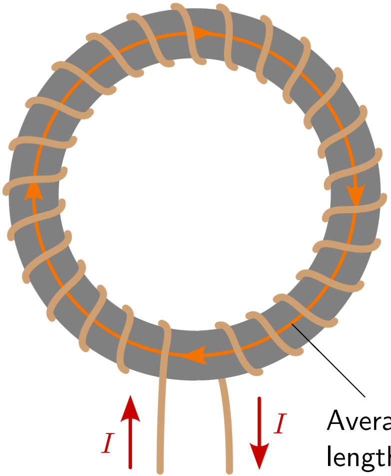
$$\Theta = \oint_s \vec{H} \cdot \mathrm{d}\vec{s} \qquad [\mathrm{A}]$$

H is the magnetic field strength
s is the edge of the area A

Magnetic field strength



Average field line length ℓ_m

If $\vec{H}$ is constant over the integration path $\mathrm{d}\vec{s}$:

$$\oint_s \vec{H} \cdot \mathrm{d}\vec{s} = |\vec{H}| \cdot \ell_\mathrm{m}$$

$$|\vec{H}| = \frac{\Theta}{\ell_\mathrm{m}} \qquad \left[\frac{\mathrm{A}}{\mathrm{m}}\right]$$

Example: Magnetic field strength

A toroidal coil with 1000 turns and an average field line length of 50 cm is traversed by a current of 100 mA. How strong is the magnetic field?

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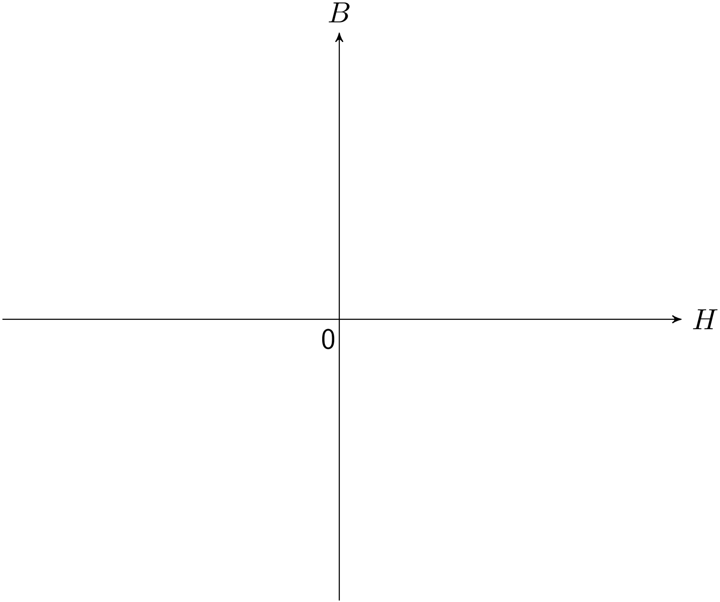
$$H = \frac{N \cdot I}{\ell_{\mathrm{m}}} = \frac{1 \cdot 50\,\mathrm{A}}{2\pi \cdot 0{,}2\,\mathrm{m}} = 39{,}79\,\frac{\mathrm{A}}{\mathrm{m}}$$

Magnetic flux and flux density



Magnetic flux density:

$\vec{B} \quad [\mathrm{T}]$

Permeability:

$\mu \quad [1]$

$$\vec{B} = \mu_0 \cdot \mu_{\mathrm{r}} \cdot \vec{H}$$

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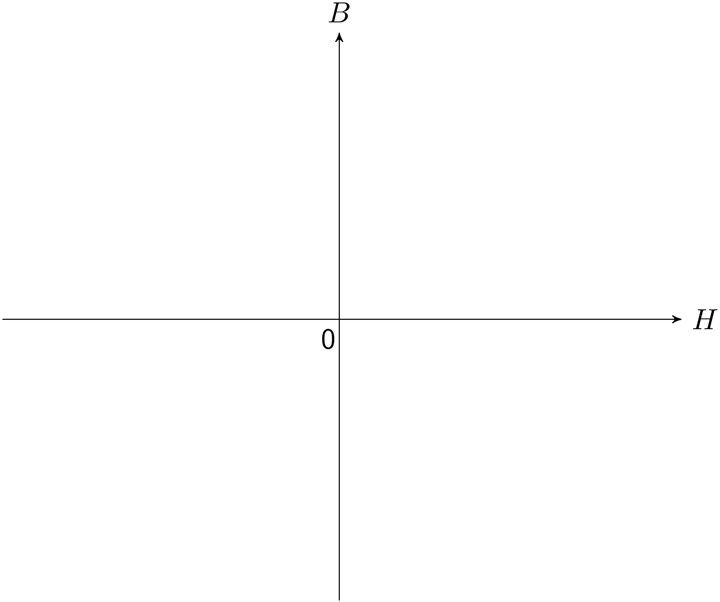

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Magnetic flux:
Φ [Wb]

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$$\vec{B} = \frac{\Phi}{\vec{A}}$$

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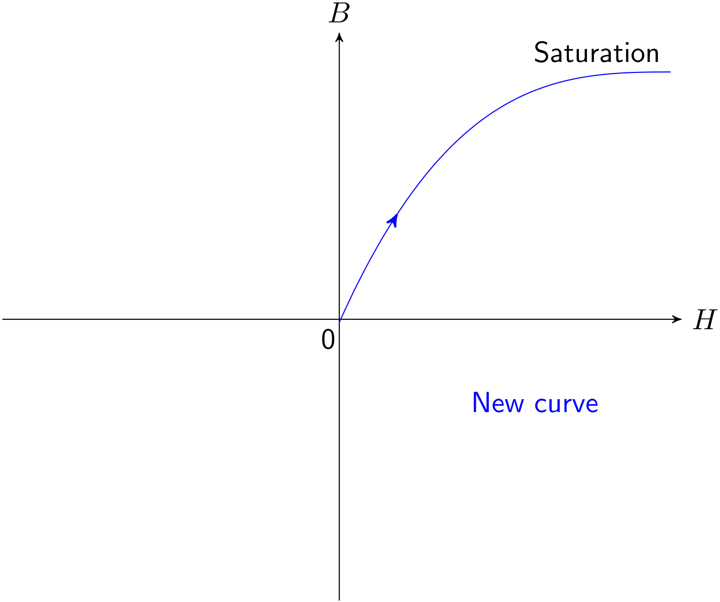

Hard magnetic:
$H_c > 10 \cdot 10^3 \, \frac{\mathrm{A}}{\mathrm{m}}$
e.g. permanent magnets

Soft magnetic:
$H_c < 500 \, \frac{\mathrm{A}}{\mathrm{m}}$
e.g. actuators

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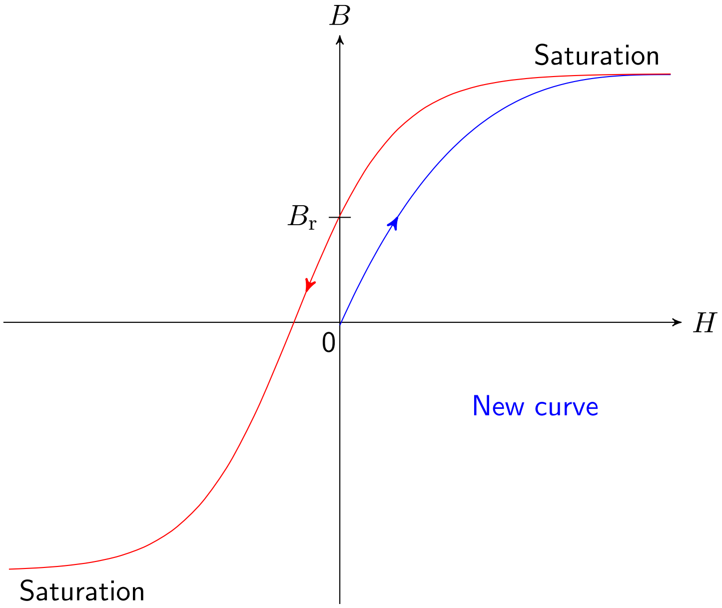

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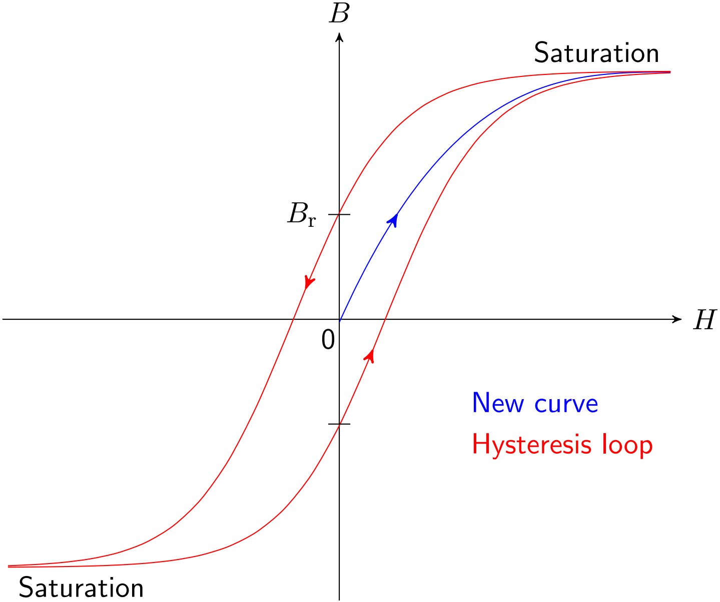
Remanence flux density
B_r



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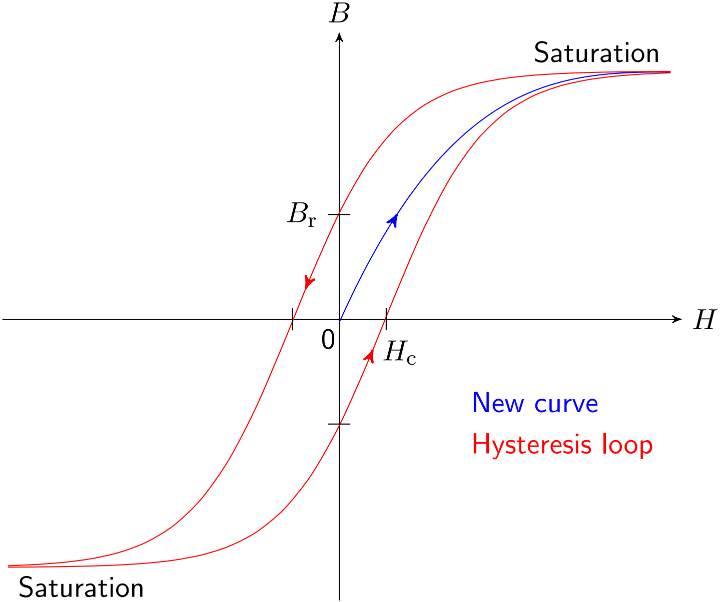

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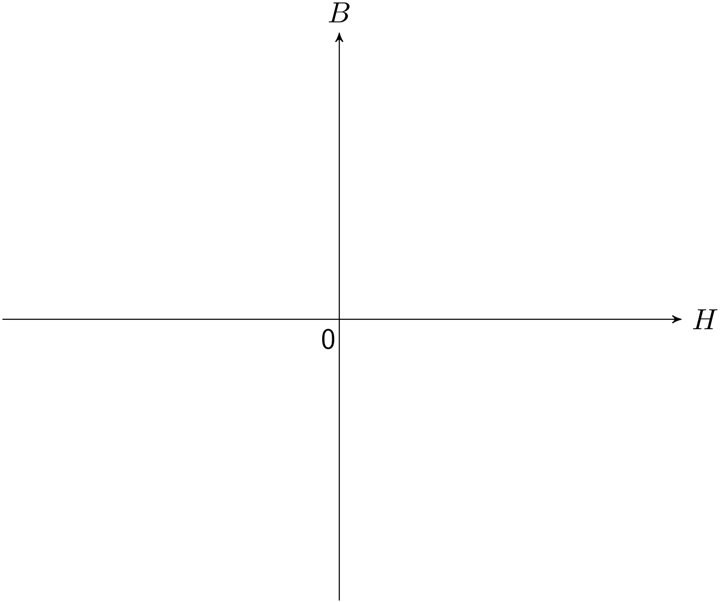
$$B_\mathrm{Eisen} = 2000 \cdot B_\mathrm{Luft} = 251{,}2\,\mathrm{mT}$$

Magnetic flux and flux density



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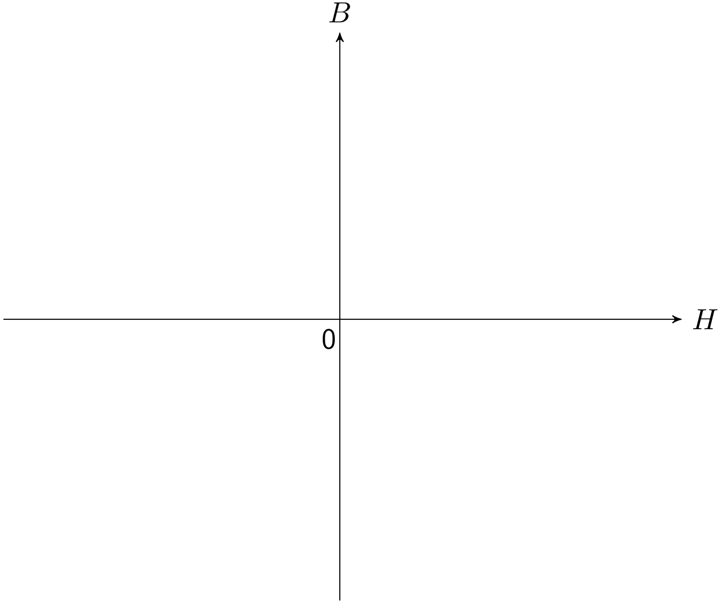

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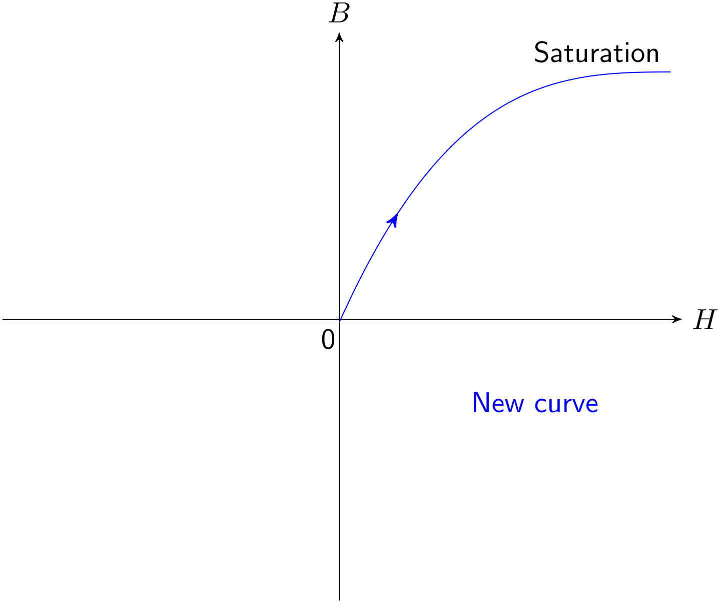

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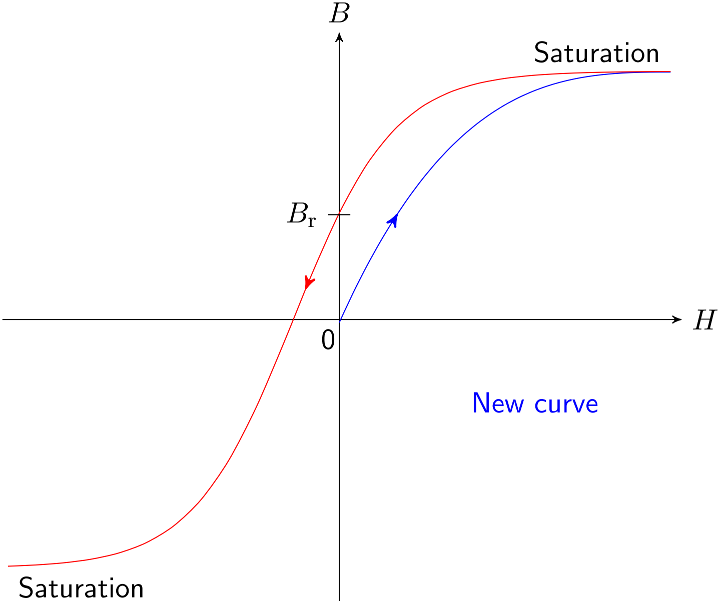

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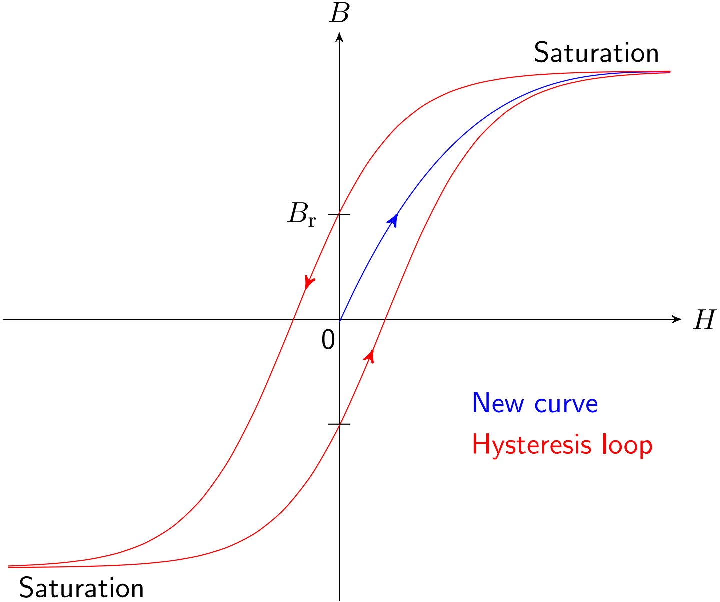

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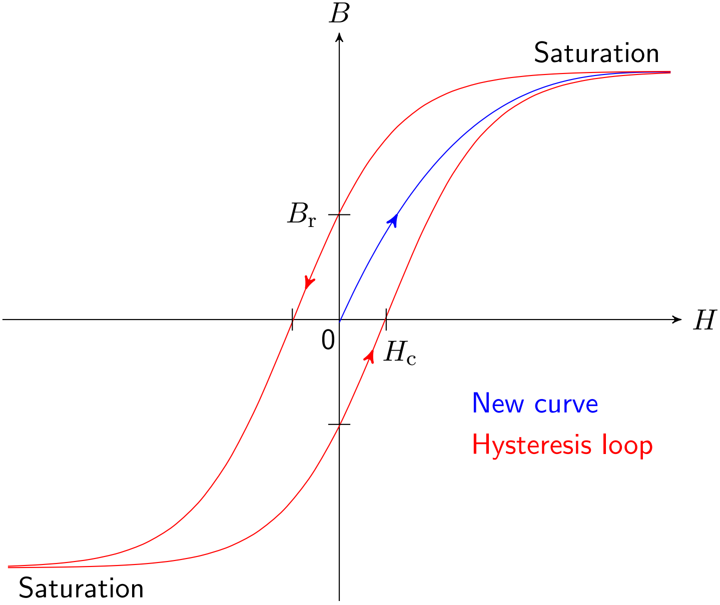

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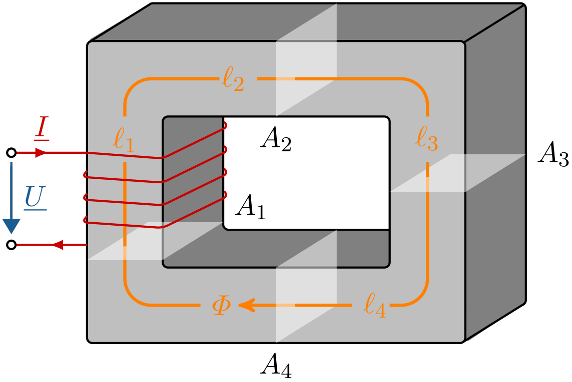
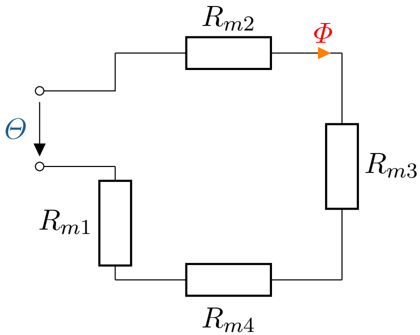
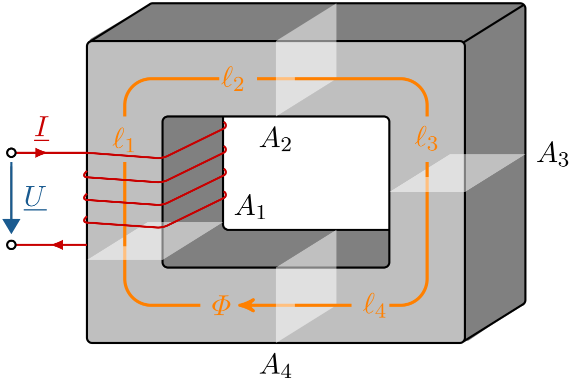
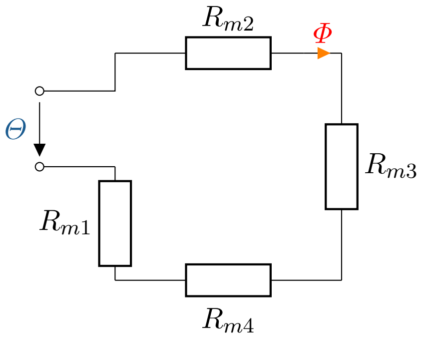
Magnetic resistance







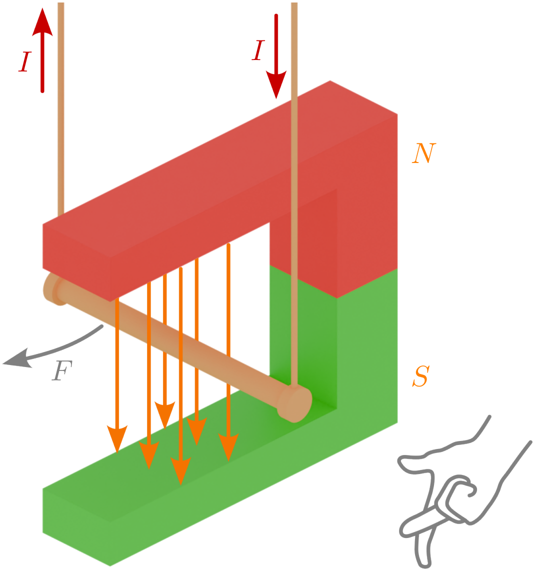


$$R_\mathrm{m} = \frac{\ell_\mathrm{m}}{\mu_\mathrm{r} \cdot \mu_0 \cdot A} \qquad \left[\frac{\mathrm{A}}{\mathrm{V} \cdot \mathrm{s}}\right]$$

Magnetic circuit

Analogous to Ohm's law for electricity, the following applies to magnetic circuits:

$$\Theta = R_{\mathrm{m}} \cdot \Phi$$


I
I
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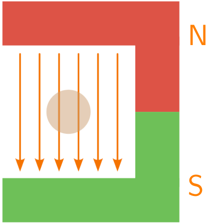
Lorentz force



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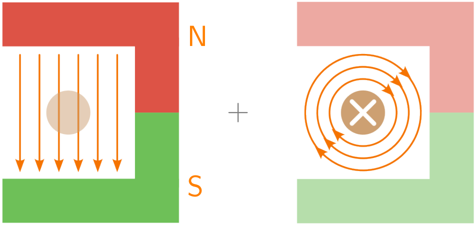

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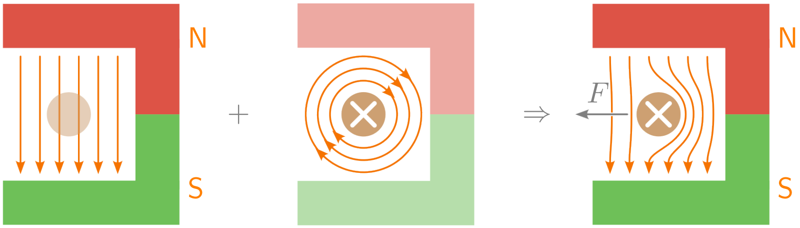

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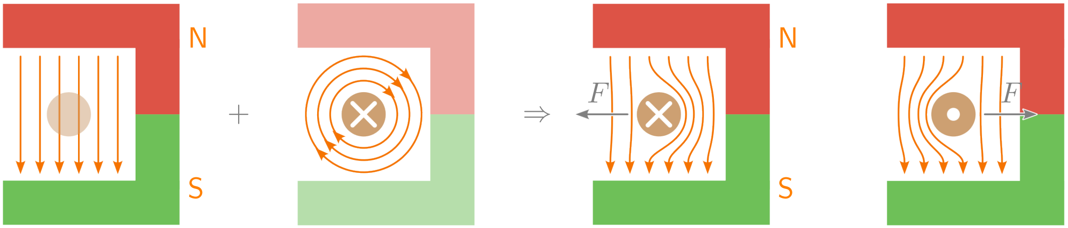
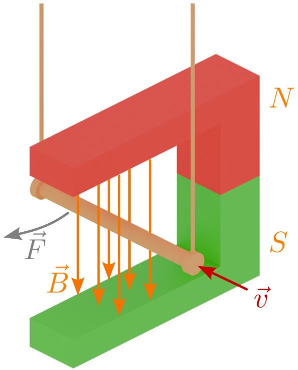


Lorentz force

A force acts on a unit charge q moving at velocity $\vec{v}$ in a magnetic field $\vec{B}$:

$$\vec{F} = q \cdot (\vec{v} \cdot \vec{B})$$



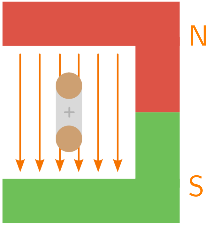
Lorentz force

$$F = q \cdot v \cdot B \cdot \sin\alpha$$

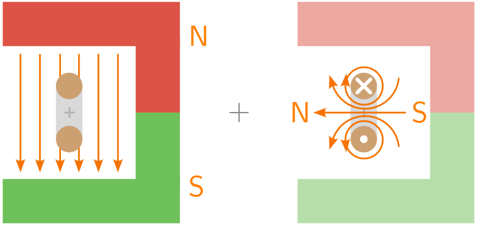
$$F = I \cdot \ell \cdot N \cdot B \cdot \sin\alpha$$

Lorentzkraft

Lorentzkraft



Lorentzkraft



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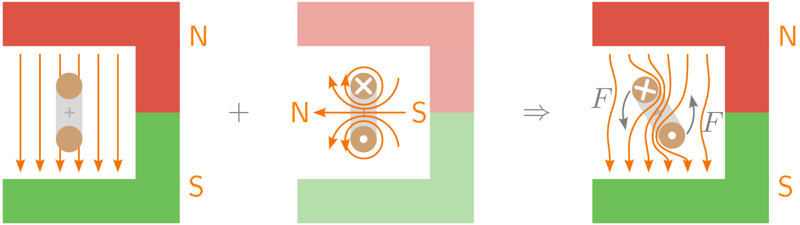
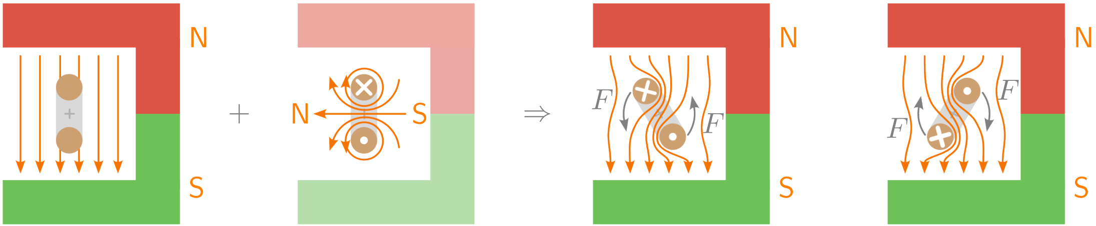


Lorentzkraft



Example: Lorentz force

A direct current motor has a magnetic flux density of $B = 0.8\,\mathrm{T}$ in the air gap. There are a total of $N = 400$ windings under the poles, through which a current of $I = 10\,\mathrm{A}$ flows. The effective conductor length is $\ell = 150\,\mathrm{mm}$.
Calculate the force F at the circumference of the anchor.

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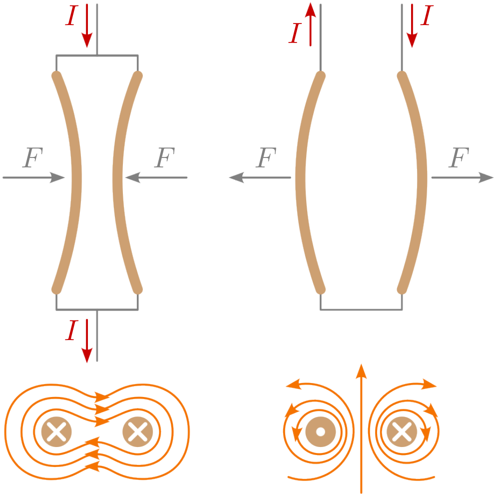
Calculate the force F at the circumference of the anchor.

$$
\begin{aligned}
F &= B \cdot I \cdot \ell \cdot N \\
&= 0{,}8\,\tfrac{\mathrm{Vs}}{\mathrm{m}^2} \cdot 10\,\mathrm{A} \cdot 0{,}15\,\mathrm{m} \cdot 400 \\
&= 480\,\frac{\mathrm{kg} \cdot \mathrm{m}^2 \cdot \mathrm{s} \cdot \mathrm{A} \cdot \mathrm{m}}{\mathrm{s}^3 \cdot \mathrm{A} \cdot \mathrm{m}^2} = 480\,\mathrm{N}
\end{aligned}
$$

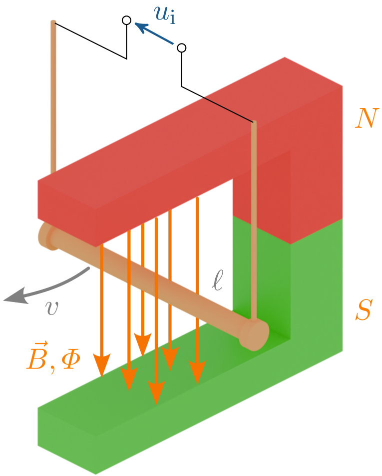
Lorentzkraft

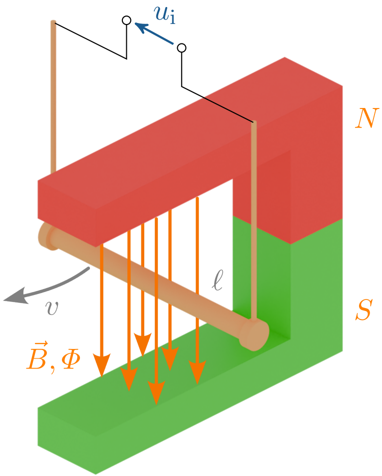
Force between two straight, parallel, thin and infinitely long conductors:

$$F_{12} = \frac{\ell \cdot \mu_0 \cdot I_1 \cdot I_2}{2 \cdot \pi \cdot r}$$



Induction





$$u_\mathrm{i} = B \cdot \ell \cdot v \cdot N$$

The magnitude of the induced voltage depends on:

- the magnetic flux density B
- the length of the conductor in the magnetic field ℓ
- the speed of movement or change in flow v
- the number of conductors in the magnetic field N

Induction

General law of induction:

$$u_{\mathrm{i}} = -N \cdot \frac{\mathrm{d}\Phi}{\mathrm{d}t}$$

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Lenz's law:

Cause and effect are always opposite.

Inductance

$$L = \frac{N \cdot \Phi}{I} \qquad [\mathrm{H}]$$

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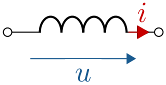
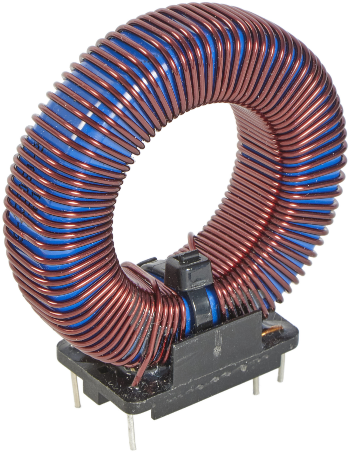


Determination of inductance



For simple geometric structures, the inductance can be determined via the magnetic resistance:

$$L = \frac{N^2}{R_\mathrm{m}}$$

Method 1 using field sizes:

1. Assumption of a current $I \rightarrow$ flow $\Theta = N \cdot I$
2. Calculate field strength $H = \frac{\Theta}{\ell_\mathrm{m}}$
3. Calculate flux density $B = \mu_\mathrm{r} \cdot \mu_0 \cdot H$
4. Determine magnetic flux $\Phi = \iint\limits_A \vec{B} \cdot \mathrm{d}\vec{A}$
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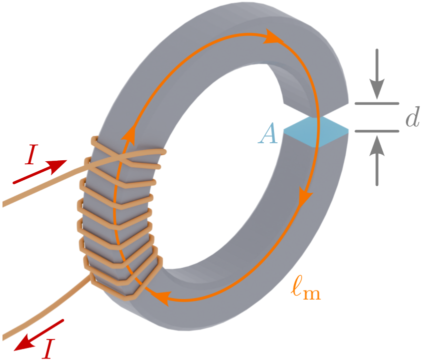
Method 2 via magnetic resistance:

1. Divide the magnetic circuit into sections and magnetic resistance R_{m} calculate $R_{\mathrm{m}} = \frac{\ell_{\mathrm{m}}}{\mu_{\mathrm{r}} \cdot \mu_0 \cdot A}$
2. Calculate total resistance (for series structure) $R_{m,\mathrm{ges}} = \sum R_{\mathrm{m}}$
3. Inductance $L = \frac{N^2}{R_{m,\mathrm{ges}}}$

Example: Inductance

For a coil with an iron core and air gap, the influence of the air gap d on the inductance L is to be investigated. The coil has $N = 100$ turns. The relative permeability is $\mu_r = 2000$. The average iron core length is $\ell_m = 5\,\mathrm{cm}$. The cross-sectional area is $A = 1\,\mathrm{cm}^2$.



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$$R_\mathrm{m} = \frac{\ell_\mathrm{m}}{\mu_\mathrm{r} \cdot \mu_0 \cdot A}$$

$$L = \frac{N^2}{R_\mathrm{m}}$$

$$L = \frac{100^2 \cdot 2000 \cdot 1{,}256 \cdot 10^{-6}\,\frac{\mathrm{Vs}}{\mathrm{Am}} \cdot 1 \cdot 10^{-4}\,\mathrm{m}^2}{0{,}05\,\mathrm{m}}$$

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L &= \frac{100^2 \cdot 2000 \cdot 1{,}256 \cdot 10^{-6}\,\frac{\mathrm{Vs}}{\mathrm{Am}} \cdot 1 \cdot 10^{-4}\,\mathrm{m}^2}{0{,}05\,\mathrm{m}} \\
L &= 0{,}05\,\frac{\mathrm{Vs}}{\mathrm{A}} = 50\,\mathrm{mH}
\end{aligned}
$$

Example: Inductance

- The air gap is now $d = 1\,\mathrm{mm}$. How large is the inductance?

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R_\mathrm{m} &= R_{m,\mathrm{Fe}} + R_{m,\mathrm{L}} \\
&= \frac{\ell_\mathrm{m} - d}{\mu_\mathrm{r} \cdot \mu_0 \cdot A} + \frac{d}{\mu_0 \cdot A}
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&= \frac{0{,}05\,\mathrm{m} - 0{,}001\,\mathrm{m}}{2000 \cdot 1{,}256 \cdot 10^{-6}\,\frac{\mathrm{Vs}}{\mathrm{Am}} \cdot 1 \cdot 10^{-4}\,\mathrm{m}^2} + \frac{0{,}001\,\mathrm{m}}{1{,}256 \cdot 10^{-6}\,\frac{\mathrm{Vs}}{\mathrm{Am}} \cdot 1 \cdot 10^{-4}\,\mathrm{m}^2}
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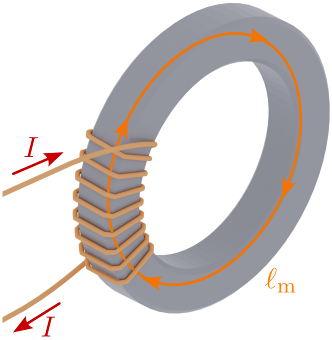
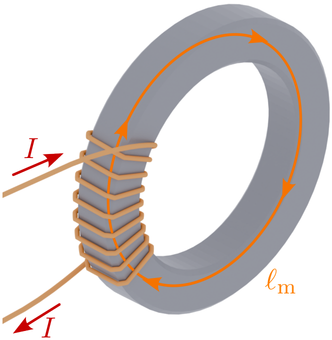
L &= 1{,}226\,\mathrm{mH}
\end{aligned}
$$

Energy in the magnetic field



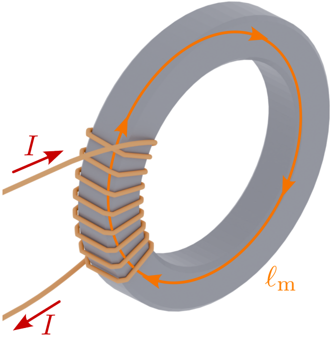
$$p = \frac{\mathrm{d}W}{\mathrm{d}t} = u \cdot i \qquad u = L \cdot \frac{\mathrm{d}i}{\mathrm{d}t}$$



$$p = \frac{\mathrm{d}W}{\mathrm{d}t} = u \cdot i \qquad u = L \cdot \frac{\mathrm{d}i}{\mathrm{d}t}$$

$$\mathrm{d}W_{\mathrm{m}} = u_{\mathrm{L}} \cdot i_{\mathrm{L}} \cdot \mathrm{d}t = L \cdot \frac{\mathrm{d}i}{\mathrm{d}t} \cdot i_{\mathrm{L}} \cdot \mathrm{d}t = L \cdot i_{\mathrm{L}} \cdot \mathrm{d}i_{\mathrm{L}}$$

Energy in the magnetic field



$$p = \frac{\mathrm{d}W}{\mathrm{d}t} = u \cdot i \qquad u = L \cdot \frac{\mathrm{d}i}{\mathrm{d}t}$$

$$\mathrm{d}W_\mathrm{m} = u_\mathrm{L} \cdot i_\mathrm{L} \cdot \mathrm{d}t = L \cdot \frac{\mathrm{d}i}{\mathrm{d}t} \cdot i_\mathrm{L} \cdot \mathrm{d}t = L \cdot i_\mathrm{L} \cdot \mathrm{d}i_\mathrm{L}$$

$$W_\mathrm{m} = L \cdot \int_0^I i_\mathrm{L} \cdot \mathrm{d}i_\mathrm{L} = \frac{1}{2} \cdot L \cdot I^2$$

$$= \frac{1}{2} \cdot N \cdot \varPhi \cdot I$$

Energy in the magnetic field

Calculation based on field sizes:

$$\begin{aligned} W_\mathrm{m} &= \ell_\mathrm{m} \cdot A \cdot \int\limits_0^{B_L} H \mathrm{d}B \\ &= \ell_\mathrm{m} \cdot A \cdot \frac{B_\mathrm{L}^2}{2 \cdot \mu_\mathrm{r} \cdot \mu_0} \end{aligned} \tag{1}$$

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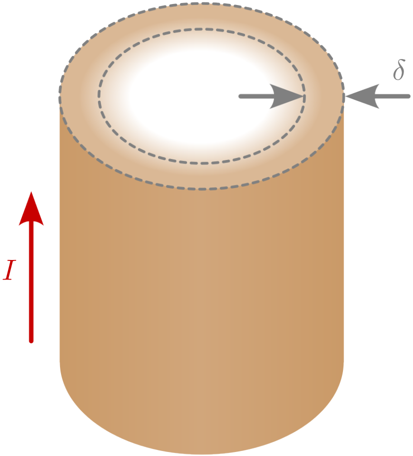
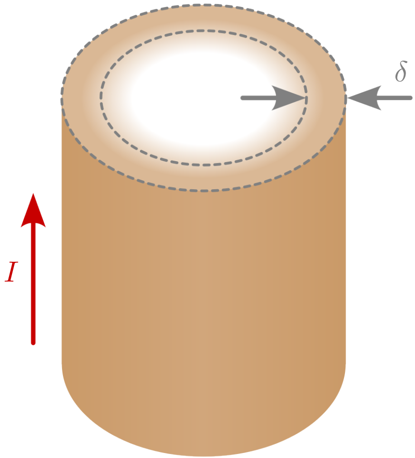
$$\frac{W_\mathrm{m}}{V} = \frac{B_\mathrm{L}^2}{2 \cdot \mu_0} = \frac{F}{A}$$

Skin-effect



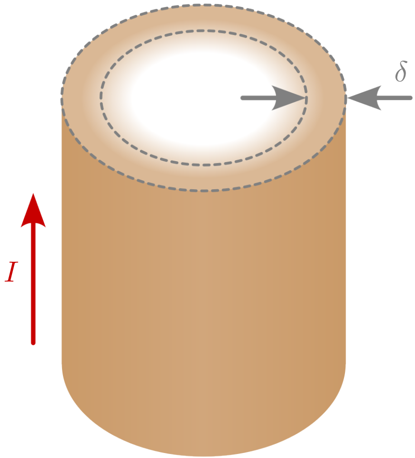
Alternating current only utilises the surface of a conductor.

Skin-effect



Alternating current only utilises the surface of a conductor.

$$\delta = \sqrt{\frac{\rho_\mathrm{R}}{\pi \cdot \mu \cdot f}}$$



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Frequency	**Skin-Depth** δ_{Cu}
$5\,\mathrm{Hz}$	$29{,}7\,\mathrm{mm}$
$50\,\mathrm{Hz}$	$9{,}38\,\mathrm{mm}$
$500\,\mathrm{Hz}$	$2{,}97\,\mathrm{mm}$
$500\,\mathrm{kHz}$	$93{,}8\,\mu\mathrm{m}$

Example: Skin-depth

A copper conductor is traversed by a current with a frequency of $f = 50\,\mathrm{Hz}$. The following values are given for the conductor:

- Absolute magnetic permeability: $\mu = 4\pi \cdot 10^{-7}\,\frac{\mathrm{Vs}}{\mathrm{Am}}$
- Specific resistance: $\rho_R = 0{,}01721\,\frac{\Omega\cdot\mathrm{mm}^2}{\mathrm{m}}$

Calculate the skin depth δ of the conductor.

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$$\delta = \sqrt{\frac{0{,}01721\,\frac{\Omega\cdot\mathrm{mm}^2}{\mathrm{m}}}{\pi \cdot 4\pi \cdot 10^{-7}\,\frac{\mathrm{Vs}}{\mathrm{Am}} \cdot 50\,\mathrm{Hz}}}$$

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Calculate the skin depth δ of the conductor.

$$\delta = \sqrt{\frac{\rho_\mathrm{R}}{\pi \cdot \mu \cdot f}}$$

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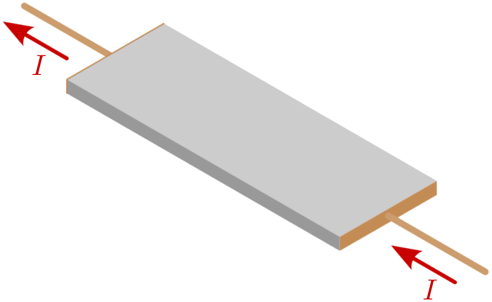
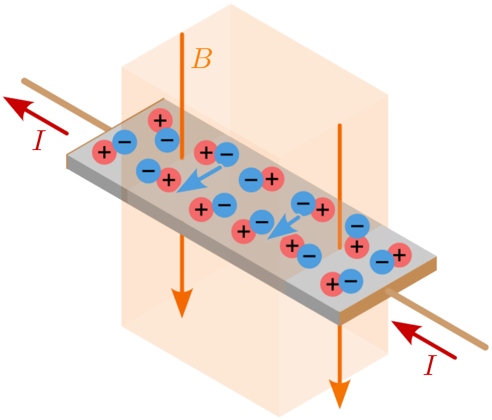
$$\delta \approx 9{,}34\,\mathrm{mm}$$

Hall-effect

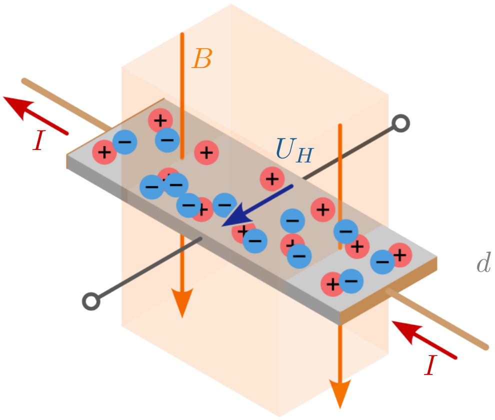


- A current I flows through a conductor plate.

Hall-effect



- A current I flows through a conductor plate.
- The Lorentz force deflects the moving charge carriers to the edge.



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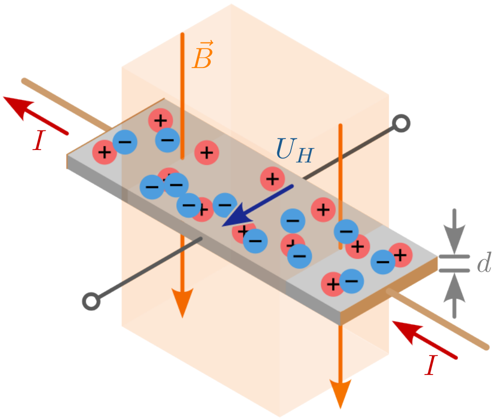

- A current I flows through a conductor plate.
- The Lorentz force deflects the moving charge carriers to the edge.
- The charge displacement generates an electric field.
- This is measured as the Hall voltage U_{H}.

$$U_{\mathrm{H}} = A_{\mathrm{H}} \cdot \frac{I \cdot B}{d}$$

Example: Hall voltage

A metal plate with a thickness of $d = 1\,\mathrm{cm}$ and a current of $I = 2\,\mathrm{A}$ is located in a magnetic field with a flux density of $B = 0.5\,\mathrm{T}$. The Hall constant of the material is $A_\mathrm{H} = 3.2 \cdot 10^{-3}\,\frac{\mathrm{m}^3}{\mathrm{C}}$. Calculate the Hall voltage.

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$$U_\mathrm{H} = 0{,}32\,\mathrm{V}$$